Handheld Console Comparisons: Lateral Consumer Machines as Musical Instruments

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Abstract

The genre of micromusic, also known as chiptune, highlights the practice of utilising obsolete videogame consoles in the composition and production of music to an extent that is beyond a novelty or simply a passing fad. This is often realised via relatively old hardware that is not designed for the composer or musician, executing modern-day music performance or composition software. These videogame consoles have thus become "lateral machines" – devices that have outlived their original purpose and have been reincarnated as musical instruments and composition environments.

Prominent examples of hardware / software combinations used in chiptune composition and performance include packages such as the Midines *(Kann, 2005) and the* SynthCart*. The* Midines *is a hardware-based data interface used to control the Nintendo Entertainment System's 2A03 sound chip via MIDI. The SynthCart allows an Atari 2600 user to control the console's various unique sounds in real time via a set of keypad input controllers.*

The GameBoy music community is active and vibrant within the micromusic genre. GameBoy hardware is a topic that has been discussed a number of times. In particular, the question of which GameBoy model has 'the best' sound is a common theme. Abstract, ill-defined terms such as 'warmth', 'retro' and 'feel' are often used in the context of this discussion. Words that are less difficult to define such as 'noise' are also used without a large amount of empirical evidence.

The aim of this paper is to investigate, clarify and extend upon recent comparisons within the field regarding the sonic qualities of various models of the Nintendo GameBoy. A number of important points and areas of interest are discussed. Sonic qualities are compared and contrasted (by means of spectral and waveform analysis) in terms of their relationship to multiple units of the same model as well as the difference between various models. In addition, the amount and type of discrepancy in quality of different units of the same model (when compared between various models) is also a noteworthy one.

This line of inquiry is based upon the sampling of a large number of GameBoy units across all models including the original GameBoy, GameBoy Color, GameBoy Pocket, GameBoy Advance and GameBoy Advance SP. To some degree, a link between serial numbers and sound quality can also be made.

This comparison focuses on the point of a consumer-targeted machine such as the Nintendo GameBoy having been developed as a toy, with apparently looser parameters for the quality of production on a unit-by-unit basis than more professional musical equipment. By providing an in-depth analysis, the GameBoy music community can be well informed about the differences between the hardware models and the sounds that they create – an important aspect when considering the philosophy of authenticity within the micromusic genre.

References

Kann, Chris. Midines. Software and hardware. Published by Wayfar. http://www.wayfar.net/index.php. 2005. Accessed March 10, 2007.

Slocum, Paul. Synthcart. Software. Published by AtariAge. http://qotile.net/synth.html. Accessed March 10, 2007.